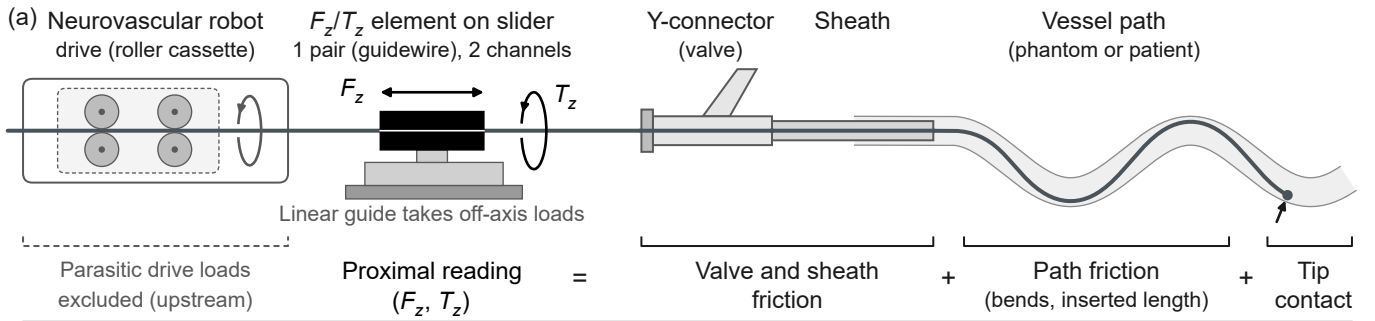
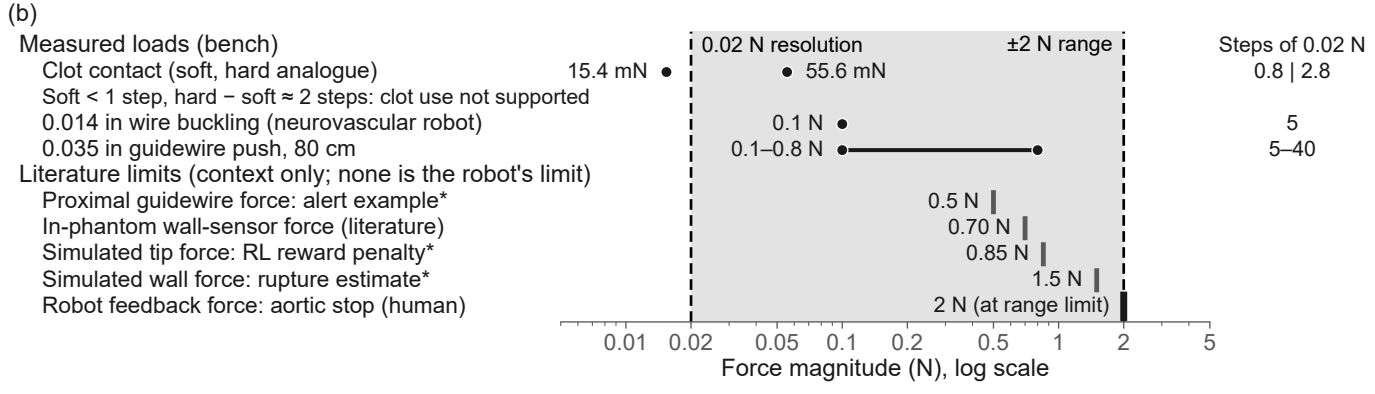

(a)
Neurovascular robot
drive (roller cassette)
F_z/T_z element on slider
1 pair (guidewire), 2 channels
F_z
T_z
Linear guide takes off-axis loads
Y-connector
(valve)
Sheath
Vessel path
(phantom or patient)
Parasitic drive loads
excluded (upstream)
Proximal reading
(F_z, T_z)
=
Valve and sheath
friction
+
Path friction
(bends, inserted length)
+
Tip
contact
Proximal reading − per-run friction baseline (valve-only and per-segment traces) = F_z excess → robot limits (set from bench baselines)


(b)
Measured loads (bench)
Clot contact (soft, hard analogue)
Soft < 1 step, hard − soft ≈ 2 steps: clot use not supported
0.014 in wire buckling (neurovascular robot)
0.035 in guidewire push, 80 cm
Literature limits (context only; none is the robot's limit)
Proximal guidewire force: alert example*
In-phantom wall-sensor force (literature)
Simulated tip force: RL reward penalty*
Simulated wall force: rupture estimate*
Robot feedback force: aortic stop (human)
0.02 N resolution
±2 N range
15.4 mN
55.6 mN
0.1 N
0.1–0.8 N
0.5 N
0.70 N
0.85 N
1.5 N
2 N (at range limit)
Steps of 0.02 N
0.8 | 2.8
5
5–40
0.01
0.02
0.05
0.1
0.2
0.5
1
2
5
Force magnitude (N), log scale